Embodiment × Minimalism

UBISS 2026 Workshop A

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Outline

- What is embodiment?
- How does embodiment relate to minimalism in robotics?
- A light smorgasbord of (hopefully inspiring) examples

Embodiment

Embodiment is a state of having a physical body.

An *embodied agent* has meaningful sensory-motor interactions with the environment.

Embodied Intelligence

- General field encompassing human embodied experiences, study of other living organisms, robotics, dance, etc.

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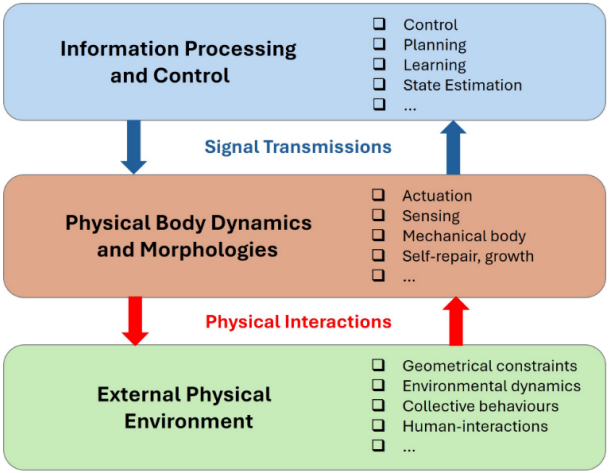
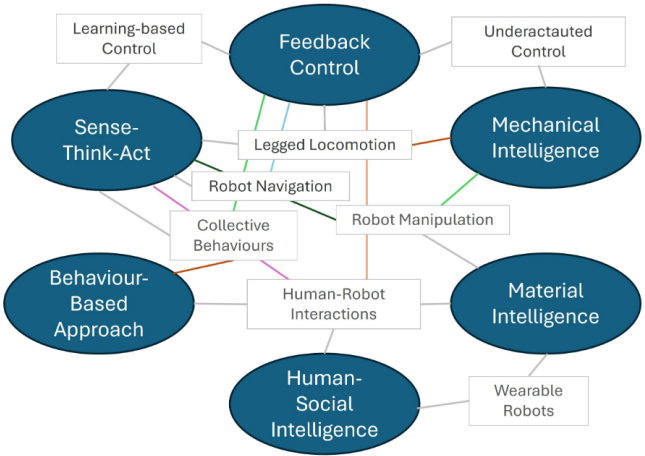
- EI conference: https://embodied-intelligence.org/
- Understanding embodied intelligence helps us shift information processing from the "brain" to the body
- Reduces explicit computational requirements

Embodied Intelligence



Fumiya Iida, Chapa Sirithunge, Perla Maiolino and Josie Hughes (2025), "Foundations of Embodied Intelligence for Robotic Systems", Foundations and Trends in Robotics

Embodied Control

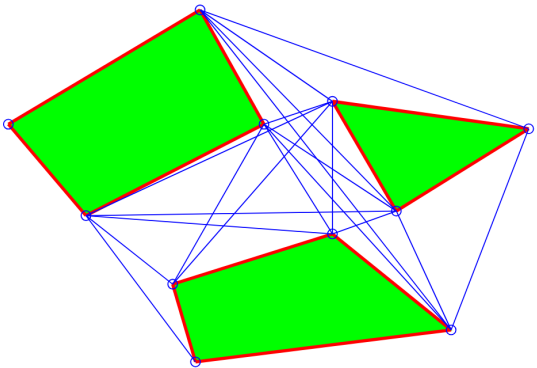


Fumiya Iida, Chapa Sirithunge, Perla Maiolino and Josie Hughes (2025), "Foundations of Embodied Intelligence for Robotic Systems", Foundations and Trends in Robotics

Embodiment and Minimalism in Robotics

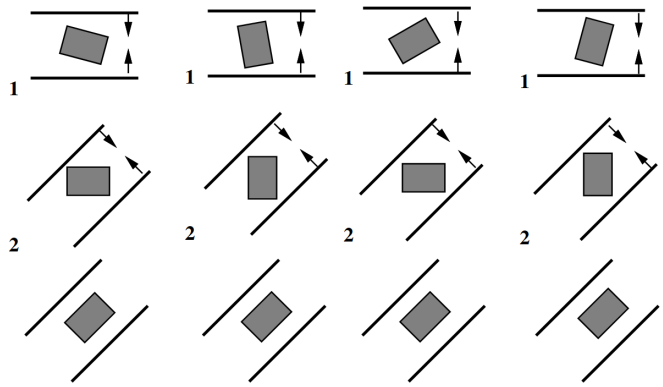
Geometry

Visibility and Reachability



Geometry

Compliance, Manipulation, Friction / Stiction



Source: Goldberg, Ken. Orienting polygonal parts without sensors. Algorithmica, 1993.

Geometry

Bouncing robots (external video)

Dynamics

- Funnel chaining in control theory

Dynamics

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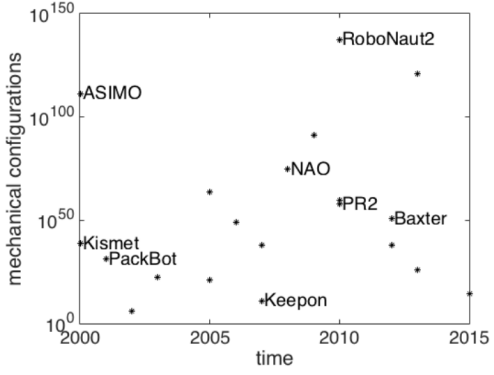
- Oscillations and fixed points

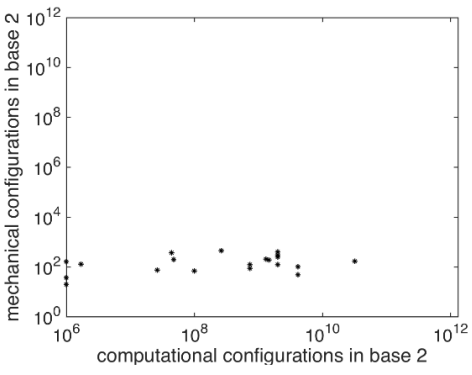
Dynamics

- Funnel chaining in control theory
- Oscillations and fixed points
- Passive Dynamic Walkers

Expressivity

Information theoretic measure of available static and dynamic states of the body





Expressivity: Embodied Example

More Concepts and Examples

Proprioception vs. Exteroception

- Proprioception: sensing of “internal” states

Proprioception vs. Exteroception

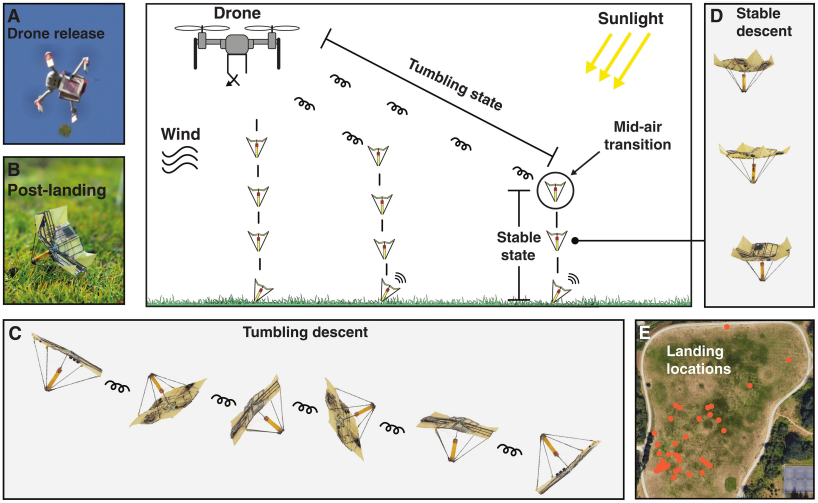
- Proprioception: sensing of “internal” states
- Exteroception: sensing the external world

Proprioception vs. Exteroception

- Proprioception: sensing of “internal” states
- Exteroception: sensing the external world
- Let’s try it!

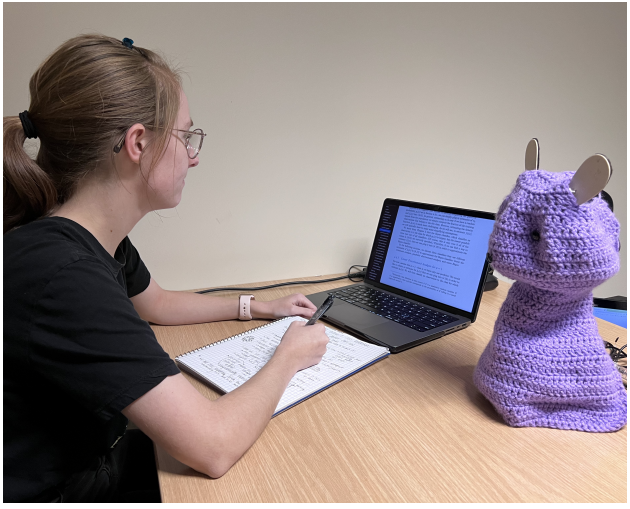
Embodied Wallfollowing

Minimal Directed Flight



Source: Johnson K, Arroyos V, Ferran A, Villanueva R, Yin D, Elberier T, Aliseda A, Fuller S, Iyer V, Gollakota S. Solar-powered shape-changing origami microfliers. Science Robotics 2023.

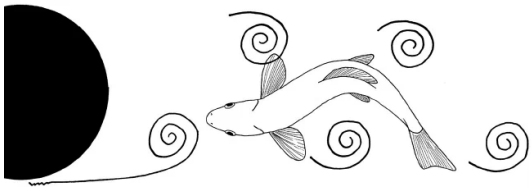
Human Response to Embodied Agents



Source: Maja Matarić, Interaction Lab

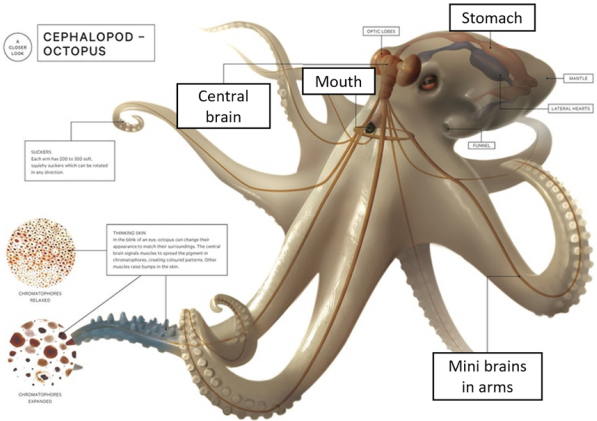
Soft Robots: Two examples

- passive fishes
- foambots (external video)



Source: Swimming Strategies for Energy Economy, Frank E. Fish

Distributed Information Processing



Collective Embodied Intelligence

Brazil Nut Effect

Collective Embodied Intelligence

Collective Braitenberg Vehicles (external video)

A Collective Embodiment Exercise

Conclusion

There is so much

- Could have whole week workshop on this topic

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- Instead, discussion: connections to docking?